

What is Amazon GuardDuty?

Amazon GuardDuty is a threat detection service that continuously monitors, analyzes, and processes AWS data sources and logs in your AWS environment. GuardDuty uses threat intelligence feeds, such as lists of malicious IP addresses and domains, file hashes, and machine learning (ML) models to identify suspicious and potentially malicious activity in your AWS environment. The following list provides an overview of potential threat scenarios that GuardDuty can help you detect:

- Compromised and exfiltrated AWS credentials.
- Data exfiltration and destruction that can lead to a ransomware event. Unusual patterns of login events in the supported engine versions of Amazon Aurora and Amazon RDS databases, that indicate anomalous behavior.
- Unauthorized cryptomining activity in your Amazon Elastic Compute Cloud (Amazon EC2) instances and container workloads.
- Presence of malware in your Amazon EC2 instances and container workloads, and newly uploaded files in your Amazon Simple Storage Service (Amazon S3) buckets.
- Operating system-level, networking, and file events that indicate unauthorized behavior on your Amazon Elastic Kubernetes Service (Amazon EKS) clusters, Amazon Elastic Container Service (Amazon ECS) - AWS Fargate tasks, and Amazon EC2 instances and container workloads.

The following video provides an overview of how GuardDuty helps you detect threats in your AWS environment.

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Features of GuardDuty

Here are some of the key ways in which Amazon GuardDuty can help you monitor, detect, and manage potential threats in your AWS environment.

Continuously monitors specific data sources and event logs

- **Foundational threat detection** – When you enable GuardDuty in an AWS account, GuardDuty automatically starts ingesting the foundational data sources associated with that account. These data sources include AWS CloudTrail management events, VPC flow logs (from Amazon EC2 instances), and DNS logs. You don't need to enable anything else for GuardDuty to start

analyzing and processing these data sources to generate associated security findings. For more information, see [GuardDuty foundational data sources](#).

- **Extended Threat Detection** – This capability detects multi-stage attacks that span foundational data sources, multiple types of AWS resources, and time, within an AWS account. There might be multiple events in your account that, individually, don't present themselves as a clear threat. However, when these events are observed in a sequence that is indicative of a suspicious activity, GuardDuty identifies it as an attack sequence. GuardDuty notifies you by generating the associated attack sequence finding type to provide details about the observed attack sequence.

With no additional cost associated with it, Extended Threat Detection is automatically enabled for each AWS account when they enable GuardDuty. This capability doesn't require you to enable any use-case focused protection plan. However, to increase the breadth of security to your Amazon S3 resources, GuardDuty recommends enabling S3 Protection in your account. This will help Extended Threat Detection to identify multi-stage attacks that potentially impact your Amazon S3 resources.

For more information about how this capability works and what threat scenarios it covers, see [GuardDuty Extended Threat Detection](#).

- **Investigation (Preview)** – GuardDuty Investigation provides AI-powered security analysis of your GuardDuty findings. You can create investigations to analyze specific findings, accounts, or your entire organization, and receive structured investigation summaries with risk assessment, confidence scoring, MITRE ATT&CK® classification, and actionable next steps. For more information, see [GuardDuty Investigation \(Preview\)](#).
- **Use-case focused GuardDuty protection plans** – For enhanced threat detection visibility into the security of your AWS environment, GuardDuty offers dedicated protection plans that you can choose to enable. Protection plans help you monitor logs and events from other AWS services. These sources include EKS audit logs, RDS login activity, Amazon S3 data events in CloudTrail, EBS volumes, Runtime Monitoring across Amazon EKS, Amazon EC2, and Amazon ECS-Fargate, Lambda network activity logs, and AWS CloudTrail data events from Amazon Bedrock, Amazon Bedrock AgentCore, and Amazon SageMaker AI for AI Protection. GuardDuty consolidates these log and event sources under the term - [Features](#). You can enable one or more dedicated protection plans in a supported AWS Region at any time. GuardDuty will start monitoring, processing, and analyzing the activities based on which protection plan you enable. For more information about each protection plan and how it works, see the corresponding protection plan document.

When you enable GuardDuty for the first time, GuardDuty automatically enables all protection plans except Runtime Monitoring for your account, and includes them in the 30-day free trial. You can disable any protection plan at any time. If you are an existing GuardDuty customer, a protection plan that launches after you enabled GuardDuty is not enabled automatically, and you can choose to enable it. For information about which protection plans are enabled by default, see [Using GuardDuty 30-day free trial](#).

Protection plan	Description
AI Protection	Detects threats to your AI workloads that use Amazon Bedrock, Amazon Bedrock AgentCore, and Amazon SageMaker AI, such as anomalous model invocations and cost harvesting attacks.
S3 Protection	Identifies potential security risks such as data exfiltration and destruction attempts in your Amazon S3 buckets.
EKS Protection	EKS Audit Log Monitoring analyzes Kubernetes audit logs from your Amazon EKS clusters for potentially suspicious and malicious activities.
Runtime Monitoring	Monitors and analyzes operating system-level events on your Amazon EKS, Amazon EC2, and Amazon ECS (including AWS Fargate), to detect potential runtime threats.
Malware Protection for EC2	Detects potential presence of malware by scanning the Amazon EBS volumes associated with your Amazon EC2 instances. There is an option to use this feature on-demand .
Malware Protection for S3	Detects potential presence of malware in the newly uploaded objects within your Amazon S3 buckets. There is an option to use this feature on-demand.

Protection plan	Description
Malware Protection for AWS Backup	Detects potential presence of malware by scanning backup resources, including EBS Snapshots, EC2 AMI's, and AWS Backup recovery points. There is an option to use this feature on-demand.
RDS Protection	Analyzes and profiles your RDS login activity for potential access threats to the supported Amazon Aurora and Amazon RDS databases.
Lambda Protection	Monitors Lambda network activity logs, starting with VPC flow logs, to detect threats to your AWS Lambda functions . Examples of these potential threats include cryptomining and communicating with malicious servers.

Enable Malware Protection for S3 independently

GuardDuty offers flexibility to use Malware Protection for S3 independently, without enabling the Amazon GuardDuty service. For more information about getting started with only Malware Protection for S3, see [GuardDuty Malware Protection for S3](#). To use all other protection plans, you must enable the GuardDuty service.

Manage multiple-account environment

You can manage a multiple-account AWS environment by using either AWS Organizations (recommended) or legacy invitation method. For more information, see [Multiple accounts in GuardDuty](#).

Generates security findings for detected threats

When GuardDuty detects potential security threats associated with your AWS resources, it starts generating security findings that provide information about the potentially compromised resource. After you enable GuardDuty in your account, generate [Sample findings](#) to view the associated [Finding details](#). For a complete list of security findings, see [GuardDuty finding types](#).

With GuardDuty, you can also use a tester script that generates specific GuardDuty security findings to understand how to review and respond to GuardDuty findings. For more information, see [Test GuardDuty findings in dedicated accounts](#).

Assessing and managing security findings

GuardDuty consolidates your security findings across accounts and displays results in the Summary dashboard on the GuardDuty console. You can also retrieve findings through the AWS Security Hub CSPM API, AWS Command Line Interface, or AWS SDK. With a holistic view of your current security status, you can identify trends and potential issues, and take necessary remediation steps. For more information, see [Managing GuardDuty findings](#).

Integrate with related AWS security services

To further help you analyze and investigate the security trends in your AWS environment, consider using the following AWS security-related services in combination with GuardDuty.

- **AWS Security Hub CSPM** – This service gives you a comprehensive view of the security state of your AWS resources and helps you check your AWS environment against security industry standards and best practices. It does this partly by consuming, aggregating, organizing, and prioritizing your security findings from multiple AWS services (including Amazon Macie) and supported AWS Partner Network (APN) products. Security Hub CSPM helps you analyze your security trends and identify the highest priority security issues across your AWS environment.

For information about using GuardDuty and Security Hub CSPM together, see [Integrating GuardDuty with AWS Security Hub CSPM](#). To learn more about Security Hub CSPM, see the [AWS Security Hub User Guide](#).

- **Amazon Detective** – This service helps you analyze, investigate, and quickly identify the root cause of security findings or suspicious activities. Detective automatically collects log data from your AWS resources. It then uses machine learning, statistical analysis, and graph theory to generate visualizations that help you to conduct faster and more efficient security investigations. The Detective prebuilt data aggregations, summaries, and context help you analyze and determine the nature and extent of potential security issues.

For information about using GuardDuty and Detective together, see [Integrating GuardDuty with Amazon Detective](#). To learn more about Detective, see the [Amazon Detective User Guide](#).

- **Amazon EventBridge** – This service helps you receive notifications and respond to GuardDuty security findings in near-real time. GuardDuty creates an event when there is a change in the findings. You can choose how frequently you want to receive the notifications from EventBridge. For more information, see [What is Amazon EventBridge](#) in the *Amazon EventBridge User Guide*.

PCI DSS Compliance

GuardDuty supports the processing, storage, and transmission of credit card data by a merchant or service provider, and has been validated as being compliant with Payment Card Industry (PCI) Data Security Standard (DSS). For more information about PCI DSS, including how to request a copy of the AWS PCI Compliance Package, see [PCI DSS Level 1](#).

For more information, see [New third-party test compares Amazon GuardDuty to network intrusion detection systems](#) in the *AWS Security Blog*.

Pricing in GuardDuty

This section focuses on the AWS Free Tier model that GuardDuty uses for various protection plans, and how you can view estimated and actual usage costs. If you are looking for the pricing details associated with all the protection plans across supported Regions, see [GuardDuty pricing](#).

AWS Free Tier

AWS Free Tier helps you explore and try out AWS services free of charge up to specified limits for each service. There are three categories – 12 months free, always free, and short-term free trials. Amazon GuardDuty belongs to the short-term free trial category and offers a 30-day free trial. When you continue using GuardDuty after this free trial ends, you start incurring cost based on how you use this service.

¹Exception to GuardDuty 30-day free trial

On-demand malware scan (under Malware Protection for EC2) and Malware Protection for S3 don't fall into the GuardDuty 30-day short term free trial category. Malware Protection for S3 includes a Free Tier whereas the On-demand malware scan follows a pay-as-you-use cost model. There is no 30-day free trial or a Free Tier cost model with On-demand malware scan.

Using GuardDuty 30-day free trial

When using GuardDuty for the first time in an AWS Region, your AWS account is automatically enrolled in a 30-day free trial in that Region. Some of the protection plans will also get enabled automatically and are included in the 30-day free trial. Because GuardDuty is a regional service, when you enable it for the first time in a different Region, your account will get a 30-day free trial of GuardDuty in that Region. When working with multiple accounts in a GuardDuty organization, each account gets its own 30-day free trial.

Use the following table to review which protection plans are enabled by default with GuardDuty, and their free trial availability.

Protection plan	Enabled by default with GuardDuty	Separate free trial availability ²	
EKS Protection	Yes	Yes	
S3 Protection	Yes	Yes	
Runtime Monitoring	No	Yes	
Malware Protection for EC2 – GuardDuty-initiated malware scan	Yes	Yes	
Malware Protection for EC2 – On-demand malware scan in GuardDuty	No	No ¹	
GuardDuty Malware Protection for S3	No	No ¹	
RDS Protection	Yes	Yes	
Lambda Protection	Yes	Yes	
Malware Protection for AWS Backup	No	No	

²When you enable GuardDuty for the first time, protection plans (except Runtime Monitoring) are automatically enabled and included in the initial 30-day free trial. When an existing GuardDuty account enables a new protection plan after their initial GuardDuty free trial has expired, then that protection plan comes with its own 30-day free trial. For more information about free trials for protection plans, see the document associated with each protection plan.

View estimated usage cost during free trial – During 30-day free trial of GuardDuty and potentially a protection plan, GuardDuty provides estimated usage cost for your account. If you're a delegated GuardDuty administrator account, you can view the total estimated usage cost and account-level breakdown for all the member accounts that have enabled GuardDuty. For more information, see [Monitoring GuardDuty Usage and Estimating Costs](#).

Usage cost after free trial ends – When you continue using GuardDuty or any of its protection plans after the free trial ends, you will start incurring associated usage costs. To view your bill, navigate to **Cost Explorer** in the <https://console.aws.amazon.com/costmanagement/> console. For more information about AWS account billing, see the [AWS Billing User Guide](#).

Using Malware Protection for S3 with Free Tier

Malware Protection for S3 includes a Free Tier associated with your AWS accounts. For more information, see [Pricing and usage cost for Malware Protection for S3](#).

Accessing GuardDuty

Amazon GuardDuty is available in most AWS Regions. For a list of Regions where GuardDuty is currently available, see [Regions and endpoints](#).

You can use GuardDuty in any of the following ways:

GuardDuty console

<https://console.aws.amazon.com/guardduty/>

The console is a browser-based interface to access and use GuardDuty. The GuardDuty console provides access to your GuardDuty account, data, and resources.

AWS Command Line Interface

With AWS Command Line Interface (AWS CLI), you can issue commands at your system's command line to perform GuardDuty tasks and AWS tasks. The AWS CLI commands are useful if you want to build scripts that perform tasks.

For information about installing and using AWS CLI, see [AWS Command Line Interface User Guide](#). To view the available AWS CLI commands for GuardDuty, see [AWS CLI Command Reference](#).

GuardDuty HTTPS API

You can access GuardDuty and AWS programmatically by using the GuardDuty HTTPS API, which lets you issue HTTPS requests directly to the service. For more information, see the [Amazon GuardDuty API Reference](#).

AWS SDKs

AWS provides software development kits (SDKs) that consist of libraries and sample code for various programming languages and platforms (Java, Python, Ruby, .NET, iOS, Android, and more). The SDKs provide a convenient way to create programmatic access to GuardDuty. For information about the AWS SDKs, including how to download and install them, see [Tools for Amazon Web Services](#).